

# **ANS PHARMACOLOGY**

## **MBS230-P&T**

### **UNIT 2.4**

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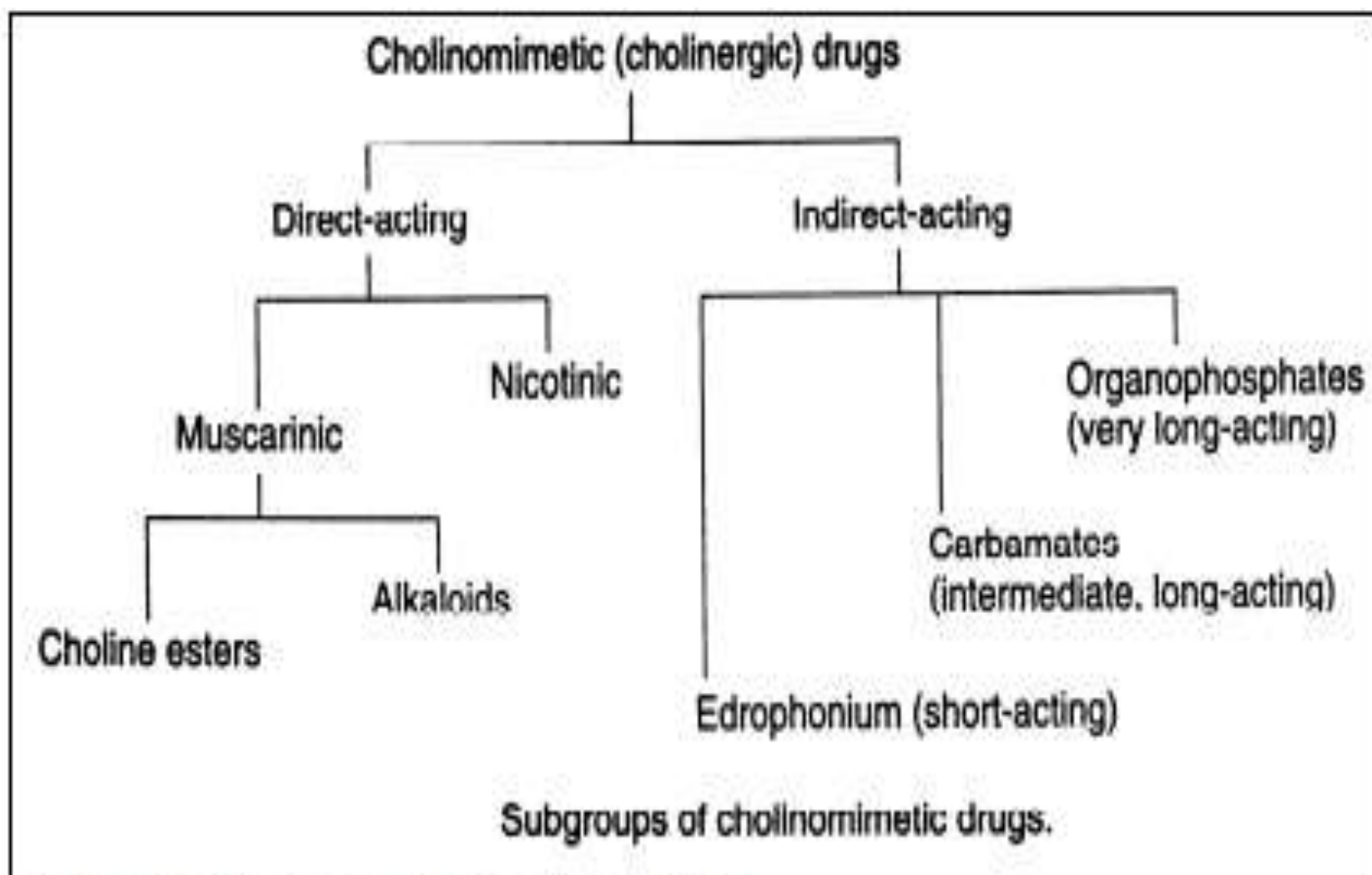
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## **2.4 MUSCARINIC RECEPTOR AGONISTS & ANTI-CHOLINESTERASES (CHOLINERGIC AGONISTS)**

# MUSCARINIC RECEPTOR AGONISTS AND ANTICHOLINESTERASES

## LS - 2.4: LEARNING OBJECTIVES:

- Describe the pharmacological effects, clinical uses and unwanted effects of muscarinic receptor agonists and anticholinesterases
- Explain why muscarinic receptor agonists and anticholinesterases have limited clinical uses in practice



**Figure 10:** Subgroups of cholinomimetic drugs.

# Introduction to Cholinergic Agonists

Two Ways to Achieve Cholinergic Effects:

1. Direct-Acting (Muscarinic Receptor Agonists)
  - Directly activate muscarinic receptors.
  - Mimic the effects of parasympathetic nervous system stimulation.
  - Also known as direct-acting parasympathomimetics.
  
2. Indirect-Acting (Anticholinesterases)
  - Inhibit the enzyme acetylcholinesterase (AChE).
  - This increases the concentration of endogenous acetylcholine (ACh) at synapses.
  - ACh then acts on both muscarinic and nicotinic receptors.
  - Also known as indirect-acting parasympathomimetics.

# Why Not Just Use Acetylcholine?

Acetylcholine (ACh) itself is rarely used clinically due to:

- Rapid hydrolysis in the GI tract (ineffective orally).
- Rapid metabolism in the blood (very short duration of action).

Fortunately, a number of congeners with resistance to hydrolysis:

methacholine: Resistant to AChE

carbachol: Potent, has both muscarinic and nicotinic

bethanechol: overwhelmingly high muscarinic and minimal nicotinic.

- There are also several other naturally occurring muscarinic agonists such as muscarine and pilocarpine.

# **DIRECT ACTING PARASYMPATHOMIMETIC DRUGS**

Mechanism: Act directly bind to muscarinic receptors and activate them

Resulting effects: Mimic many of the physiological effects that result from stimulation of the parasympathetic division of the autonomic nervous system

# MUSCARINIC AGONISTS: PHARMACOLOGICAL EFFECTS

| Effector           | Effects                                                                                                                                          |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Heart              | Negative chronotropic effect and decreased conduction velocity                                                                                   |
| Arterioles         | Vasodilatation<br>Blood pressure falls due to a fall in total peripheral resistance                                                              |
| Lacrimation        | Increases                                                                                                                                        |
| Respiratory system | Bronchoconstriction<br>Increased bronchial secretions                                                                                            |
| GIT                | Increased peristaltic activity and motility<br>Increased salivation and GIT secretions                                                           |
| Urinary tract      | Increased contraction of the ureter and bladder smooth muscle (detrusor)<br>Increased trigone and sphincter relaxation<br>(promotes micturition) |
| Sweat glands       | Increased sweat gland secretion                                                                                                                  |
| Eye                | Miosis and accommodation<br>Increased outflow of aqueous humour resulting in a reduction in intraocular pressure                                 |
| Other effects      | Tremor, ataxia                                                                                                                                   |

# Pharmacological Effects Cont'

**Table 2: Muscarinic Autonomic Effects of Acetylcholine (\*clinically important)**

|                                      |                                       |
|--------------------------------------|---------------------------------------|
| <b>*Iris sphincter muscle</b>        | <b>Contraction (miosis)</b>           |
| <b>*Ciliary muscle</b>               | <b>Contraction (near vision)</b>      |
| <b>*SA node</b>                      | <b>Bradycardia</b>                    |
| <i>Atrium</i>                        | <b>Reduced contractility</b>          |
| <b>*AV node</b>                      | <b>Reduced conduction velocity</b>    |
| <i>Arteriole</i>                     | <b>Dilation (via nitric oxide)</b>    |
| <b>*Bronchial muscle</b>             | <b>Contraction</b>                    |
| <b>*Gastrointestinal motility</b>    | <b>Increased</b>                      |
| <b>*Gastrointestinal secretion</b>   | <b>Increased</b>                      |
| <i>Gallbladder</i>                   | <b>Contraction</b>                    |
| <b>*Bladder (detrusor)</b>           | <b>Contraction</b>                    |
| <b>*Bladder (trigone, sphincter)</b> | <b>Relaxation</b>                     |
| <i>Penis</i>                         | <b>Erection (but not ejaculation)</b> |
| <i>Sweat glands</i>                  | <b>Secretion</b>                      |
| <i>Salivary glands</i>               | <b>Secretion</b>                      |
| <i>Lacrimal glands</i>               | <b>Secretion</b>                      |
| <i>Nasopharyngeal glands</i>         | <b>Secretion</b>                      |

# 1. Bethanechol

## USES

- Gastroparesis: **stimulates** gastrointestinal motility and secretion.
- Urinary Retention (Bethanechol): Non-obstructive
- Xerostomia: dry mouth, nasal passages, and throat occurring in Sjögren's syndrome

Adverse Effects: Abdominal cramps, hypotension, bradycardia.  
Rarely, myocardial ischemia (in high doses).

## 2. Pilocarpine

Uses:

- Open-angle glaucoma (topical). Lowers intraocular pressure rapidly.
- Xerostomia (more commonly than bethanechol).
- Adverse Effects: Miosis (blurred vision), increased sweating, nausea, bradycardia, bronchospasm.

### **3. Methacholine**

- Use: Bronchoprovocation test for diagnosing asthma.
- Inhaled to elicit bronchoconstriction. A significant drop in lung function indicates airway hyperreactivity.

: used for treatment of glaucoma when pilocarpine is ineffective (topical)

## 4. Carbachol

Use: Topical treatment for glaucoma (used when pilocarpine is ineffective).

Contraindications for Direct-Agonists:

- Peptic ulcer disease
- Bronchial asthma
- Cardiac disease (bradyarrhythmias)
- Parkinson's disease (can worsen symptoms)

# Cholinesterase Inhibitors

Mechanism of Action (MOA):

- Inhibit acetylcholinesterase (AChE), the enzyme that breaks down ACh.
- This leads to an accumulation of ACh at all cholinergic synapses.

Effects:

- Muscarinic: Same as direct-acting agonists.
- Nicotinic:
  - Skeletal Muscle: Increased strength of contraction (beneficial in myasthenia gravis).
  - Autonomic Ganglia: Complex effects.
- CNS: Variable, depending on ability to cross the blood-brain barrier.

# Classification based on duration and mechanism of action:

| Class                             | Duration      | Examples                                                            | Key Features                                                                                                                                                                                      |
|-----------------------------------|---------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Short-acting</b>               | 2-10 min      | <b>Edrophonium</b>                                                  | Used for the <b>diagnosis</b> of myasthenia gravis.                                                                                                                                               |
| <b>Intermediate-acting</b>        | 2-6 hrs       | <b>Neostigmine</b><br><b>Physostigmine</b><br><b>Pyridostigmine</b> | <b>Neostigmine:</b> Poor CNS penetration.<br><b>Physostigmine:</b> Crosses the BBB, used for anticholinergic toxicity.<br><b>Pyridostigmine:</b> Mainstay <b>treatment</b> for myasthenia gravis. |
| <b>Long-acting (Irreversible)</b> | Days to weeks | <b>Organophosphates</b> (e.g., Ecothiophate, Parathion, Sarin)      | Form a covalent bond with AChE. Used as insecticides and nerve gases                                                                                                                              |

# Anticholinesterases: Clinical Uses

Despite their broad effects, they have specific clinical applications:

1. Myasthenia Gravis (MG)
  - Pathology: Autoimmune disease with antibodies against nicotinic receptors at the neuromuscular junction.
  - Treatment: Pyridostigmine (oral, intermediate-acting). Increases ACh to activate remaining receptors.
  - Diagnosis: Edrophonium (short-acting IV). A transient improvement in muscle strength confirms the diagnosis.

# Anticholinesterases: Clinical Uses

## 2. Glaucoma

- Agents: Physostigmine, Ecothiophate (topical).
- Effect: Increase aqueous humor outflow by contracting ciliary muscle and opening trabecular meshwork.

## 3. Reversal of Neuromuscular Blockade

- Agent: Neostigmine (post-operatively).
- Effect: Reverses non-depolarizing muscle relaxants used during surgery.

# ORGANOPHOSPHATE POISONING

- Mechanism: Irreversible phosphorylation of AChE. This leads to a massive accumulation of ACh, causing a cholinergic crisis.

## Clinical Features (Toxicity):

- Muscarinic Effects: DUMBBELSS
  - Diarrhea, Urination, Miosis (pinpoint pupils), Bradycardia, Bronchospasm, Emesis, Lacrimation, Salivation, Sweating.
- **Nicotinic Effects:** Muscle fasciculations, weakness, paralysis.
- **CNS Effects:** Confusion, convulsions, coma.
- **Cause of Death:** Respiratory paralysis (due to CNS depression, muscle paralysis, and bronchospasm).

# **ORGANOPHOSPHATE POISONING ... CONT'D**

## **Treatment**

1. Supportive therapy – maintain blood pressure, patent airway, mechanical ventilation, control convulsions etc.
2. The drug of choice is atropine (muscarinic receptor antagonist); it reverses the muscarinic effects of the organophosphates

## Pharmacological Treatment

| Drug               | Action                                                                                                   | Key Point                                                                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Atropine</b>    | <b>Muscarinic antagonist.</b> Reverses muscarinic effects (e.g., bradycardia, secretions, bronchospasm). | <b>Drug of choice.</b> Does NOT reverse nicotinic effects (e.g., muscle weakness). Give in high doses until signs of <u>atropinization</u> appear (dilated pupils, dry skin, tachycardia). |
| <b>Pralidoxime</b> | <b>Cholinesterase reactivator.</b> Breaks the organophosphate-AChE bond.                                 | <b>Must be given early</b> (within minutes to hours) before "aging" of the bond occurs. Effective against nicotinic effects (e.g., muscle paralysis).                                      |
| <b>Diazepam</b>    | <b>Anticonvulsant.</b>                                                                                   | Controls seizures.                                                                                                                                                                         |

# LEARNING OUTCOMES

Students must be able to comprehend:

- Pharmacological effects, clinical uses and unwanted effects of muscarinic receptor agonists and anticholinesterases
- Why muscarinic receptor agonists and anticholinesterases have limited clinical uses

***END***

**Thanks for listening**

